Lecture 23

Analytic differential equations

Objectives. *Complex Picard iteration gives analytic solutions at ordinary points. At a regular singular point, powers replace Taylor series. We prove the nonresonant Frobenius construction and use the Gauss equation to connect series, integrals and continuation around a singular point.*

By the end of this lecture you will be able to:

- Construct a local analytic solution by a contraction.
- Prove convergence of the two nonresonant Frobenius series.
- Derive the Gauss series and its Euler integral under their parameter conditions.
- Compute the change of one logarithmic solution after continuation around a loop.

23.1 Complex Picard iteration

For a vector in $\mathbb{C}^m$ use the maximum norm. An *analytic* input $F(z, y)$ here means a locally convergent joint power series in the scalar z and the vector components of y. Convergence is absolute and uniform on smaller closed polydisks. Termwise differentiation on such polydisks follows by the same geometric majorants as for one variable in Theorem 1.5. We do not require a theorem about equivalent several-variable definitions.

Theorem 23.1 (Analytic Picard theorem). *Suppose F is analytic near $(z_0, y_0) \in \mathbb{C} \times \mathbb{C}^m$. For some $r > 0$ there is a holomorphic solution of*

$$y'(z) = F(z, y(z)), \qquad y(z_0) = y_0, \qquad |z - z_0| < r.$$

Any two holomorphic solutions with these initial data agree near z_0.

Proof. Choose a closed polydisk $|z - z_0| \le R$, $\|y - y_0\| \le b$ inside the region of convergence, with a slightly larger polydisk still inside. The differentiated series are uniformly convergent there. Consequently $\|F\| \le M$ and $\|F(z, u) - F(z, v)\| \le L\,\|u - v\|$ for finite constants M, L. The latter bound follows by integrating the derivative with respect to y along the line segment from u to v in the convex polydisk.

Choose $0 < r < R$ with $rM \le b$ and $rL < 1$. Let X consist of continuous functions on $\overline{D(z_0, r)}$ that are holomorphic inside and satisfy $\|y - y_0\|_\infty \le b$. It is a closed subset of a complete supremum-norm space by Theorem 1.1. Holomorphic functions are closed under uniform limits by Morera's theorem, Theorem 10.8, so X is complete. Define the Picard map by

$$(\mathcal{P}y)(z) = y_0 + \int_{z_0}^{z} F(\zeta, y(\zeta))\,\mathrm{d}\zeta.$$

The path is the straight segment. Composition with y is holomorphic because the joint series becomes a locally uniform series of holomorphic functions. Cauchy's primitive theorem on a disk, Theorem 9.4, shows that the integral has derivative $F(z, y(z))$. The segment length gives

$$\|\mathcal{P}y - y_0\|_\infty \le rM, \qquad \|\mathcal{P}y - \mathcal{P}v\|_\infty \le rL\,\|y - v\|_\infty .$$

The contraction principle, Theorem 1.6, gives a unique fixed point. Differentiating its integral equation proves the differential equation. Any other solution stays within the same y polydisk after shrinking its disk by continuity. Integration of its derivative makes it a fixed point on that smaller disk, where the same contraction proves uniqueness. ∎

For example, the equation $y' = y^2$ with $y(0) = 1$ has solution $y(z) = (1 - z)^{-1}$. The derivative is $(1 - z)^{-2} = y(z)^2$. Its pole at 1 shows why a local theorem does not promise an entire solution. A linear second-order equation with analytic coefficients at a point becomes a first-order system for (y, y'). Thus it has a unique analytic solution for each pair of initial values there.

23.2 A regular singular point

An equation has *at most a regular singular point* at zero if it can be written

$$z^2 y'' + z p(z) y' + q(z) y = 0, \qquad p, q \text{ holomorphic on } |z| < R. \tag{23.1}$$

This includes equations whose normalised coefficients extend to an ordinary point. Write $p(z) = \sum_{j\ge0} p_j z^j$ and $q(z) = \sum_{j\ge0} q_j z^j$. The *indicial polynomial* is $P(r) = r(r-1) + p_0 r + q_0$. A *Frobenius solution* has the form $y = z^r \sum_{n\ge0} a_n z^n$ with $a_0 \neq 0$, on a chosen branch of $z^r = \exp(r \log z)$.

Theorem 23.2 (Nonresonant Frobenius theorem). *Let r_1, r_2 be the roots of P, with $r_1 - r_2 \notin \mathbb{Z}$. On the slit disk $0 < |z| < R$, $-\pi < \arg z < \pi$, there are solutions*

$$y_j(z) = z^{r_j} A_j(z), \qquad A_j(0) = 1, \qquad j = 1, 2,$$

where A_j is holomorphic on $|z| < R$. These solutions are linearly independent and span every holomorphic solution on the slit disk.

Proof. Fix one root r and set $a_0 = 1$. Substitution and collection of the coefficient of z^{r+n} give

$$P(r+n)a_n = -\sum_{j=1}^{n} \big((r+n-j)p_j + q_j\big) a_{n-j}, \qquad n \ge 1. \tag{23.2}$$

The constant coefficient is $P(r)a_0 = 0$. Since $P(r+n)$ can vanish only if $r+n$ is the other root, the noninteger difference ensures that every denominator is nonzero. This defines all the coefficients.

We prove convergence rather than assuming the formal series is a solution. Choose $0 < u < v < R$. Cauchy's coefficient bound, Equation (10.3), gives $|p_j|, |q_j| \le M v^{-j}$ for one M. Also $P(r+n) = n^2 + (2r + p_0 - 1)n$, so $|P(r+n)| \ge cn^2$ for all integers $n \ge 1$, with some $c > 0$. If $|a_m| u^m \le B$ for $m < n$, the recurrence gives

$$|a_n| u^n \le \frac{MB(|r| + n + 1)}{cn^2} \sum_{j=1}^{n} (u/v)^j \le \frac{CB}{n}.$$

The constant C does not depend on n or B. Choose $N \geq C$ and choose B to bound the finitely many coefficients with $n \leq N$. Induction proves the bound for every n. For each $|z| < u$ the series is dominated by $B \sum (|z|/u)^n$. Since $u < R$ was arbitrary, its radius of convergence is at least R. Termwise differentiation now justifies the substitution and proves the equation.

The Wronskian, defined by $W(y_1, y_2) = y_1 y_2' - y_1' y_2$, is distinct from the flow domain W in Lecture 2. It satisfies

$$W(y_1, y_2)(z) = (r_2 - r_1) z^{r_1 + r_2 - 1} (1 + O(z))$$

near zero on the slit disk. This follows by differentiating $z^{r_j}(1+O(z))$ with analytic remainders. The coefficient $r_2 - r_1$ is nonzero. Hence the two initial-value vectors are independent at some ordinary point close to zero. For any other solution choose a linear combination with the same value and derivative at that point. Analytic Picard uniqueness gives equality near that point. The identity theorem, Theorem 10.6, on the connected slit disk gives equality everywhere. This also proves linear independence there. ∎

For the Euler equation with $p(z) = 1/2$ and $q(z) = -1/2$, the indicial polynomial is $(r - 1)(r + 1/2)$. The recurrence has zero right side, so the solutions are z and $z^{-1/2}$. Counterclockwise continuation around zero multiplies the second solution by $e^{-\pi \mathrm{i}} = -1$.

A nonconstant coefficient makes the recurrence produce genuine series. Consider

$$z^2 y'' + z(1 + z) y' - \frac{1}{16} y = 0.$$

Here $p_0 = 1$, $p_1 = 1$ and $q_0 = -1/16$; all other coefficient terms vanish. The indicial polynomial and its roots are

$$P(r) = r^2 - \frac{1}{16}, \qquad r_1 = \frac{1}{4}, \qquad r_2 = -\frac{1}{4}.$$

Their difference is not an integer. For either root the recurrence reduces to

$$a_n = -\frac{r + n - 1}{n(n + 2r)} a_{n-1}, \qquad a_0 = 1.$$

For the positive root the first steps give

$$a_1 = -\frac{1/4}{3/2} = -\frac{1}{6}, \qquad a_2 = -\frac{5/4}{5}\left(-\frac{1}{6}\right) = \frac{1}{24}, \qquad a_3 = -\frac{9/4}{21/2}\frac{1}{24} = -\frac{1}{112}.$$

The positive-root coefficients above, together with those for the negative root, give the two solutions

$$y_1(z) = z^{1/4}\left(1 - \frac{z}{6} + \frac{z^2}{24} - \frac{z^3}{112} + O(z^4)\right),$$
$$y_2(z) = z^{-1/4}\left(1 + \frac{z}{2} - \frac{z^2}{8} + \frac{7z^3}{240} + O(z^4)\right).$$

The theorem proves convergence of the full coefficient series, not just cancellation in a finite truncation. One positive circuit around zero multiplies the first solution by i and the second by −i. The analytic coefficient series themselves return unchanged.

If the root difference is an integer, the recurrence can fail at a zero denominator. The resonant Frobenius theorem is a cited input from Bender–Orszag, §3.3, pp. 68–75: order the roots so that $r_1 - r_2 = N \geq 0$. One solution is $z^{r_1} A_1(z)$. A second independent solution has the form $C y_1(z) \log z + z^{r_2} A_2(z)$, with analytic coefficient series. The logarithmic coefficient C can be zero for distinct roots. For a repeated root it is nonzero. Connection formulas in this case are not consequences of the nonresonant proof above.

23.3 Gauss's equation and Euler's integral

The *rising factorial* is $(a)_0 = 1$ and $(a)_n = a(a+1)\cdots(a+n-1)$ for integers $n \geq 1$. The *Gauss function* is initially the power series

$$ {}_2F_1(a,b;c;z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} z^n, \qquad |z|<1, \qquad c \notin \{0,-1,-2,\ldots\}. \tag{23.3} $$

Its coefficients d_n satisfy $(n+1)(n+c)d_{n+1} = (n+a)(n+b)d_n$. Unless the series terminates, their successive ratio tends to 1. Thus the series converges absolutely for $|z|<1$. The coefficient of z^n in the equation below is precisely this recurrence with all terms moved to one side. Termwise differentiation therefore proves

$$ z(1-z)y'' + [c-(a+b+1)z]y' - aby = 0. \tag{23.4} $$

For example, ${}_2F_1(1,2;3;z) = 1 + 2z/3 + z^2/2 + \cdots$.

The Gauss equation has at most regular singular points at zero and one. At zero, multiplication by $z/(1-z)$ puts it in the Frobenius form, with $p(0) = c$ and $q(0) = 0$. Its indicial polynomial is therefore

$$ P_0(r) = r(r-1) + cr = r(r+c-1). $$

The exponents at zero are 0 and $1-c$. At one use the local coordinate $\zeta = 1-z$ and put $Y(\zeta) = y(1-\zeta)$. Since $y' = -Y'$ and $y'' = Y''$, the transformed equation is

$$ \zeta(1-\zeta)Y'' + [a+b+1-c-(a+b+1)\zeta]Y' - abY = 0. $$

Its Frobenius coefficients have values $p(0) = a+b+1-c$ and $q(0) = 0$. Thus the indicial polynomial at one is

$$ \begin{aligned} P_1(r) &= r(r-1) + (a+b+1-c)r \\ &= r(r+a+b-c). \end{aligned} $$

The exponents at one are 0 and $c-a-b$.

Theorem 23.3 (Euler's integral). *If* $\operatorname{Re} c > \operatorname{Re} b > 0$, *then on* $\mathbb{C} \setminus [1,\infty)$ *the Gauss series continues to the holomorphic function*

$$ \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-zt)^{-a}\,dt. \tag{23.5} $$

Endpoint powers use real logarithms and $(1-zt)^{-a} = \exp(-a\log(1-zt))$ *uses the principal logarithm.*

Proof. For z off $[1,\infty)$ and $0 \leq t \leq 1$, the number $1-zt$ is off the nonpositive real axis. On each compact set of these z, its principal logarithm is uniformly bounded, by continuity on the compact product with $[0,1]$. Thus the modulus of the integrand is bounded by a constant times $t^{\operatorname{Re} b - 1}(1-t)^{\operatorname{Re}(c-b)-1}$. This is integrable. The parameter theorem, Theorem 10.9, proves holomorphy.

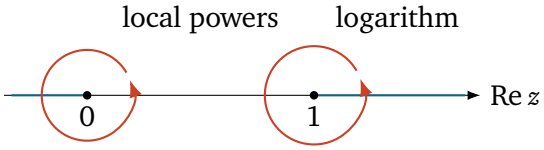
For $|z|<1$, expand $(1-zt)^{-a} = \sum_{n\geq 0}(a)_n (zt)^n/n!$ uniformly in $t \in [0,1]$. This binomial series follows by expanding the analytic function $e^{-a\log(1-w)}$: the equation $(1-w)f' = af$ recursively gives its coefficients $(a)_n/n!$, starting from $f(0) = 1$. Uniform absolute convergence allows termwise integration against the integrable endpoint weight. The Beta identity, Theorem 13.2, gives

$$ \frac{B(b+n,c-b)}{B(b,c-b)} = \frac{(b)_n}{(c)_n}. $$

The Gamma recurrence proves this equality, including complex parameters in the stated region. Multiplying the remaining coefficient gives exactly the series in Equation (23.3). This proves continuation. ∎

23.4 Continuation around a singular point

Recall from § 19.1 that a *germ* records a holomorphic function near a point. Two such functions represent the same germ if they agree on some neighbourhood of that point. Continuation of a local solution around a loop can change its germ. This change is its *monodromy* along that loop. The Frobenius factors above change by $e^{2\pi \mathrm{i} r_j}$ around zero, while their analytic coefficient functions return to their original values. The loops and the chosen cuts have different roles in this statement.



The cuts select single branches. A continuation loop crosses a cut and tracks the analytic germ rather than reassigning the principal value.

For $a = b = 1$ and $c = 2$, the exponents at zero are 0 and -1. At one both exponents are zero. This example is resonant at both points. Euler's integral, Theorem 23.3, evaluates it directly:

$$ {}_2F_1(1, 1; 2; z) = \int_0^1 \frac{\mathrm{d}t}{1 - zt} = -\frac{\log(1 - z)}{z}. $$

The apparent singularity at zero is removable, with value 1. A counterclockwise loop around 1, avoiding 0, makes $1 - z$ wind counterclockwise once around zero. Continued logarithm increases by $2\pi\mathrm{i}$. The solution therefore changes by $-2\pi\mathrm{i}/z$. This is an explicit monodromy calculation, not a claim about the full monodromy group of arbitrary Gauss parameters.

23.5 Two other equation families

The *Kummer equation* is $zy'' + (c - z)y' - ay = 0$. The *Bessel equation* of order μ is $z^2y'' + zy' + (z^2 - \mu^2)y = 0$. The order μ is distinct from the frequency ν of the Fourier convention. Both equations have at most a regular singular point at zero. Their indicial polynomials are respectively $r(r + c - 1)$ and $r^2 - \mu^2$. Thus the Frobenius mechanism applies when the root difference is nonintegral. Their global connection formulas are not treated here. In quantum radial equations the choice of a solution at one endpoint does not by itself determine its behaviour at the other.

Summary.

- An analytic joint-power-series input gives a unique local analytic IVP solution.
- Noninteger indicial-root difference gives two convergent Frobenius series. Resonant logarithmic alternatives are cited.
- The Gauss recurrence gives its differential equation. Euler's integral continues its series when $\operatorname{Re} c > \operatorname{Re} b > 0$.
- A loop changes a local power by $e^{2\pi \mathrm{i} r}$. The logarithmic Gauss example changes by $-2\pi\mathrm{i}/z$ around 1.

The turning-point analysis in Lecture 28 will use analytic model equations while retaining a separate approximation error.

References

G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, AMS, 2012, §4.1, pp. 111–114. C. M. Bender and S. A. Orszag, *Advanced Mathematical Methods for Scientists and Engineers*, McGraw–Hill, 1978, §§3.1–3.3, pp. 62–75. N. Jacob and K. P. Evans, *A Course in Analysis*, vol. IV, World Scientific, 2018, Chs. 20 and 22, pp. 335–339 and 379–382. The source uses the same Gauss parameters and Euler branch.